THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 2

Chapter 4: **Electricity: Magnetic and Heating Effects** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. An electric bell uses the: 1

(a) heating effect only **(b)** chemical effect only
(c) magnetic effect of current **(d)** light effect only

2. A magnetic compass is placed under a wire. When the switch is closed, the needle turns. This shows: 1

(a) heating effect **(b)** chemical effect
(c) light effect **(d)** magnetic effect of current

3. Which of these is an insulator? 1

(a) aluminium **(b)** silver **(c)** iron **(d)** plastic

4. A short circuit happens when: 1

(a) a cell is old **(b)** a bulb fuses
(c) live and neutral wires touch **(d)** a switch is ON

5. To make a stronger electromagnet we can: 1

(a) use a plastic core **(b)** increase the number of turns of the coil
(c) remove the iron core **(d)** reduce the current

6. Oersted found that a current-carrying wire: 1

(a) melts always
(b) has no effect near a magnet
(c) deflects a magnetic compass needle placed near it
(d) attracts only iron

7. Overloading of a circuit means: 1

(a) a broken wire **(b)** using a switch
(c) too many appliances connected at once **(d)** no current

8. A fuse is connected in: 1

(a) only in cells **(b)** with a compass
(c) series with the appliance **(d)** parallel across the appliance

9. When the current in a wire is reversed, the compass needle deflects: 1

(a) in the opposite direction **(b)** in the same direction
(c) spins **(d)** does not deflect

10. A circuit breaker (MCB) switches off the current: 1

(a) never
(b) to charge the cell
(c) only by hand
(d) automatically when current exceeds a limit

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. State the purpose of a fuse in a circuit. 2

12. What is a short circuit? 2

13. Describe Oersted's observation. 2

14. Name two devices that work on the heating effect of current. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. Describe an activity to show the magnetic effect of current. 3

16. Why do electric wires heat up when the current is large? 3

17. Explain why electrician's tools have plastic or rubber handles. 3

OR

How can an electromagnet be made? State two ways to make it stronger. 3

18. Why does a thick copper wire not heat up as much as a thin nichrome wire carrying the same current? 3

OR

Explain what happens to the compass needle when the wire carrying current is held 3
(a) above and
(b) below the needle with the same current.

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain the working of an electric bell with a labelled description. 5

OR

Describe the common electrical safety rules at home and in school. 5

20. Describe the heating effect of electric current and name devices based on it. Explain why nichrome is used. 5

OR

Describe the effect of the direction of current and of a compass position on the deflection, citing the experiment. 5

END OF QUESTION PAPER Total: 40 Marks

UNSTEPWISE